Positional Number Systems: Decimal, Binary & Hexadecimal

Data Representation & Binary Number Systems • Standard: CCSS.MATH.PRACTICE.MP5

1. MATHEMATICAL CONCEPT & STACK APPROACH: Decompose integers into powers of 2 and powers of 16 using [0] and [1] blocks on the Place Value Tray, and verify base conversions.

Binary (base-2) and hexadecimal (base-16) follow the exact same positional place-value principle as base-10, using powers of 2 (8, 4, 2, 1) or 16 instead of powers of 10. Every 4-bit binary group (0000 to 1111) maps to exactly one hexadecimal digit (0 to F).

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Place Value Studio (Binary [0]/[1] Register Trays & Base-N Columns)

Hands-On Setup: Use the sockets of the Place Value Tray as binary bit positions (8, 4, 2, 1). Place [1] Digit Blocks for active '1' bits and [0] Digit Blocks for '0' bits.

Student Prediction (Before Calculator): Place [1], [0], [1], [1] in the 8, 4, 2, 1 sockets: binary 1011_2 = 8 + 0 + 2 + 1 = 11_10. In hexadecimal (A=10, B=11), 11_10 = 0B_16.

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [1] [1] [↖] [BASE]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1] [1]	`11`	Enter decimal integer 11 in DEC mode
[↖] [BASE]	`HEX 0B`	Press [f:] [BASE]: screen displays Hexadecimal 0B!
[↖] [BASE]	`BIN 1011`	Press [f:] [BASE]: screen displays Binary 1011!
[↖] [BASE]	`OCT 13`	Press [f:] [BASE]: screen displays Octal 13!
[↖] [BASE]	`DEC 11`	Press [f:] [BASE]: returns to Decimal 11

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (Leibniz's Binary Medallion & Claude Shannon's Boolean Relay Circuits)

In 1697, Gottfried Wilhelm Leibniz designed a medallion showing that every integer can be built from 1 and 0 alone using powers of 2 (8, 4, 2, 1)—and in 1937, 21-year-old MIT student Claude Shannon proved that electrical switches wired in series (AND) and parallel (OR) execute Boolean logic directly! Using [0] and [1] blocks on the Place Value Tray, students see how place value powers modern computing.

Core Investigation (Student Warm-Up A):

Express decimal 255 as a sum of powers of 2 (128 + 64 + 32 + 16 + 8 + 4 + 2 + 1) to write its 8-bit binary and 2-digit hexadecimal representations.

Bitwise Logic: AND Masking & Column-by-Column Comparison

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1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: Bitwise logic operations (AND, OR, XOR) compare corresponding binary place-value columns of two integers in parallel. Placing value A in the top row and mask B in the bottom row lets you inspect each bit column directly.

Blocks & Prediction: Row 1 of the Place Value Tray holds nibble A ([1][1][0][0]_2 = 12). Row 0 holds mask B ([1][0][1][0]_2 = 10). Prediction: Bitwise AND: bit 3 (1 & 1 = 1), bit 2 (1 & 0 = 0), bit 1 (0 & 1 = 0), bit 0 (0 & 0 = 0). Result: 1000_2 = 8_10.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[1] [2] [ENTER] [1] [0] [AND]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1] [2]	12	Enter operand A = 12 (binary 1100)
[ENTER]	12	12 lifted to register Y
[1] [0]	10	Enter operand B = 10 (binary 1010) in X
[AND]	8	Bitwise AND evaluates 1100 & 1010 = 1000 (8 decimal)!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Confusing bitwise [AND] / [OR] with decimal addition or multiplication (e.g., thinking 12 AND 10 is 22).

Hands-On Block Remediation: Model 12 ([1][1][0][0]_2) in Row 1 and 10 ([1][0][1][0]_2) in Row 0 of the Place Value Tray using [1] and [0] Digit Blocks; only columns where BOTH rows hold [1] remain 1 in 12 AND 10 = 8 ([1][0][0][0]_2).

Misconception 2: Reading hexadecimal [10] as ten instead of decimal 16 ($1 \times 16^1 + 0 \times 16^0$).

Hands-On Block Remediation: Place [1] in the Sixteens column and [0] in the Ones column on the Place Value Tray, then switch from [HEX] [10] to [DEC] on the SC-32 to see 16.

4. EXTENSION MATHEMATICAL INSIGHT

Bitwise logic operations (AND, OR, XOR) compare corresponding binary place-value columns of two integers in parallel. Placing value A in the top row and mask B in the bottom row lets you inspect each bit column directly.

Digital Logic, Base Conversions & Computer Architecture

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: POSITIONAL NUMBER BASES & BITWISE LOGIC RULES

- Binary (Base 2) uses powers of 2 (..., 8, 4, 2, 1) with digits 0 and 1; Hexadecimal (Base 16) groups 4 binary bits into a single digit 0–9 or A–F.
- Bitwise AND (&): output bit is 1 only when BOTH input bits in that column are 1 (used for masking and network subnets).
- Bitwise OR (|): output bit is 1 when EITHER input bit is 1. Bitwise XOR (^): output bit is 1 when the two input bits DIFFER.

Key Sequence: 1 1 ↰ BASE

Investigation 1: Decompose decimal 42 into powers of 2 (32, 16, 8, 4, 2, 1) to write its 6-bit binary representation.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: Convert hexadecimal 2A_16 to decimal using place value ($2 \times 16^1 + 10 \times 16^0$).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: Evaluate the bitwise AND of 15 (1111_2) and 7 (0111_2) in binary and decimal.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: Evaluate the bitwise XOR of 12 (1100_2) and 10 (1010_2) in binary and decimal.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 12: Digital Logic, Base Conversions & Computer Architecture — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 9–12 / AP CS

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): Express decimal 255 as a sum of powers of 2 (128 + 64 + 32 + 16 + 8 + 4 + 2 + 1) to write its 8-bit binary and 2-digit hexadecimal representations.

Core Solution: Binary: 11111111_2 (eight 1s); Hexadecimal: FF_16 ($15 \times 16 + 15 = 255$).

Core Check Keys: [2] [5] [5] [↰] [BASE]

Extension Challenge (Warm-Up B): Compare 12 (1100_2) and 3 (0011_2) column by column to find the result of 12 OR 3 in both binary and decimal.

Extension Solution: 1100_2 OR 0011_2 = 1111_2 = 15_10.

Ext. Check Keys: [1] [2] [ENTER] [3] [OR]

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Decompose decimal 42 into powers of 2 (32, 16, 8, 4, 2, 1) to write its 6-bit binary representation.

Expected Answer: 32 + 8 + 2 → 101010_2

SC-32 Verification Keys: [4] [2] [↰] [BASE]

Investigation 2: Convert hexadecimal 2A_16 to decimal using place value ($2 \times 16^1 + 10 \times 16^0$).

Expected Answer: 32 + 10 = 42

SC-32 Verification Keys: [In] [HEX] [mode:] [2] [A] [->] [switch] [to] [DEC]

Investigation 3: Evaluate the bitwise AND of 15 (1111_2) and 7 (0111_2) in binary and decimal.

Expected Answer: 0111_2 = 7

SC-32 Verification Keys: [1] [5] [ENTER] [7] [AND]

Investigation 4: Evaluate the bitwise XOR of 12 (1100_2) and 10 (1010_2) in binary and decimal.

Expected Answer: 0110_2 = 6

SC-32 Verification Keys: [1] [2] [ENTER] [1] [0] [XOR]

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.